06.06.2025

BE (COMP.) / Sem-VII / R-19 / C scheme / NLP

Duration: 3hrs **Max Marks:80**

N.B. : (1) Question No 1 is Compulsory.
(2) Attempt any three questions out of the remaining five.
(3) All questions carry equal marks.
(4) Assume suitable data, if required and state it clearly.

1 Attempt any FOUR [20]

a Compare Derivational & Inflectional morphology

b What is the output of Morphological Analysis for Regular Verb, Irregular verb, Singular noun, Plural noun.

c What are the limitations of Hidden Markov Model (HMM) and MaxEnt Model for POS Tagging.

d Explain pre-processing steps generally used in NLP.

e Explain following Syntactic and Semantic Constraints on Co reference
1) Number Agreement 2) Person & Case Agreement

2 a Explain concepts of Bi-gram and n-gram with formula. [10]

For following corpus, apply Bi-gram model

Training Corpus:

<s> I am Sam </s> <s> Sam I am </s> <s> Sam I like </s>

<s> Sam I do like </s> <s> do I like Sam </s>

1. What is the most probable next word predicted by the model for the following word sequences?

(a) <s> Sam . . . (b) <s> Sam I do . . . (c) <s> Sam I am Sam . . .

(d) <s> do I like . . .

2. Which of the following sentences is better, i.e., gets a higher probability with this model?

(e) <s> Sam I do I like </s>

(f) <s> Sam I am </s>.

(g) <s> I do like Sam I am </s>

b Explain different stages of NLP. Also explain generic NLP system. [10]

QP code: 83588

prog. code: 1T00737

3 a i) Why there is need of word sense disambiguation [10]

ii) Explain Naive Bayes Supervised algorithm for Word sense Disambiguation

b Explain Shift Reduce Parser in NLP with example [10]

4 a [10]

<S>	Martin	Justin	can	watch	Will	<E>
<S>	Spot	will	watch	Martin	<E>	
<S>	Will	Justin	spot	Martin	<E>	
<S>	Martin	will	pat	Spot	<E>	

Foe given above corpus, S indicates start of the statement and E indicates end of the statement

N: Noun [Martin, Justin, Will, Spot, Pat]

M: Modal verb [can , will]

V:Verb [watch, spot, pat]

Create Transition Matrix & Emission Probability Matrix

Statement is **"Justin will spot Will"**

Apply Hidden Markov Model and do POS tagging for given statements

b How Anaphora Resolution is performed with Hobbs and Centering Algorithm [10]

5 a For a given grammar using CYK or CKY algorithm parse the statement [10]

"The man read this book" Rules:

S $\rightarrow$ NP VP	Det $\rightarrow$ *that* | *this* | *a* | *the*
S $\rightarrow$ Aux NP VP	Noun $\rightarrow$ *book* | *flight* | *meal* | *man*
S $\rightarrow$ VP	Verb $\rightarrow$ *book* | *include* | *read*
NP $\rightarrow$ Det NOM	Aux $\rightarrow$ *does*
NOM $\rightarrow$ Noun	
NOM $\rightarrow$ Noun NOM	
VP $\rightarrow$ Verb	
VP $\rightarrow$ Verb NP	

b Explain the significance of regular expression in NLP. [10]

6 Write Short Note [20]

a Explain Semi-supervised method (Yarowsky) Unsupervised (Hyperlex) [10]

b Explain Question Answering System with Algorithmic approach [10]

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(2) Attempt any three questions out of the remaining five.
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Q1a) Explain the applications of Natural Language processing. **5M**

Q1b) Illustrate the concept of tokenization and stemming in Natural Language processing. **5M**

Q1c) Discuss the challenges in part of speech tagging` . **5M**

Q1d) Describe the semantic analysis in Natural Language processing. **5M**

Q2a) Explain inflectional and derivational morphology with an example **10M**

Q2b) Illustrate the working of Porter stemmer algorithm **10M**

Q3a) Explain hidden markov model for POS based tagging. **10M**

Q3b) Demonstrate the concept of conditional Random field in NLP **10M**

Q4a) Explain the Lesk algorithm for Word Sense Disambiguation. **10M**

Q4b) Demonstrate lexical semantic analysis using an example **10M**

Q5a) Illustrate the reference phenomena for solving the pronoun problem **10M**

Q5b) Explain Anaphora Resolution using Hobbs and Cantering Algorithm **10M**

Q6a) Demonstrate the working of machine translation systems **10M**

Q6b) Explain the Information retrieval system **10M**

Paper / Subject Code: 42175 / NATURAL LANGUAGE PROCESSING (DLOC - III)

BE | Comp | Sem-VIII | CBCGS | R-19-20 | C-scheme | NLP | SH-24

30|11|2024

Duration: 3hrs **Max Marks:80**

QP-10068540

N.B. : (1) Question No 1 is Compulsory.
(2) Attempt any three questions out of the remaining five.
(3) All questions carry equal marks.
(4) Assume suitable data, if required and state it clearly.

1		Attempt any FOUR	[20]
	a	What is word sense disambiguation?	
	b	Explain reference resolution in detail	
	c	Explain rule-based machine translation systems	
	d	What is hybrid POS tagging?	
	e	Differentiate between Syntactic ambiguity and Lexical Ambiguity	
2	a	Design FST for regular and plural nouns.	[10]
	b	Explain the preprocessing operations in natural language processing	[10]
3	a	Consider the following corpus <s> a/DT dog/NN chases/V a/DT cat/NN </s> <s> the/DT dog/NN barks/V loudly/RB </s> <s> a/DT cat/NN runs/V fast/RB </s> Compute the emission and transition probabilities for a bigram HMM. Also, decode the following sentence using the Viterbi algorithm. The cat chases the dog.	[10]
	b	Compare and contrast Hobbs' Algorithm and Centering Theory.	[10]
4	a	Explain how the supervised learning approach can be applied for word sense disambiguation	[10]
	b	Explain the N-gram language model and its application.	[10]
5	a	Explain the Porter Stemming algorithm in detail.	[10]
	b	Construct a parse tree for the following sentence using the given CFG rules: The tall girl sings. Rules: S → NP VP NP → Det Adj N | Det N VP → V | V NP Det → "the" Adj → "tall" N → "girl" V → "sings"	[10]
6	a	Explain text summarization in detail	[10]
	b	Explain how Maximum Entropy is used for sequence labeling.	[10]

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Time: 3 hours **Max. Marks: 80**

N.B. (1) Question No. 1 is compulsory

(2) Assume suitable data if necessary

(3) Attempt any three questions from the remaining questions

Q.1 Solve any Four out of Five **5 marks each**

a Explain the challenges of Natural Language processing.

b Explain how N-gram model is used in spelling correction

c Explain three types of referents that complicate the reference resolution problem.

d Explain Machine Translation Approaches used in NLP.

e Explain the various stages of Natural Language processing.

Q.2 10 marks each

a What is Word Sense Disambiguation (WSD)? Explain the dictionary based approach to Word Sense Disambiguation.

b Represent output of morphological analysis for Regular verb, Irregular verb, singular noun, plural noun Also Explain Role of FST in Morphological Parsing with an example

Q.3 10 marks each

a Explain the ambiguities associated at each level with example for Natural Language processing.

b Explain Discourse reference resolution in detail.

Q.4 10 marks each

a

<S>	Martin	Justin	can	watch	Will	<E>
<S>	Spot	will	watch	Martin	<E>	
<S>	Will	Justin	spot	Martin	<E>	
<S>	Martin	will	pat	Spot	<E>	

For given above corpus,

N: Noun [Martin, Justin, Will, Spot, Pat]

M: Modal verb [can , will]

V:Verb [watch, spot, pat]

Create Transition Matrix & Emission Probability Matrix

Statement is "Justin will spot Will"

Apply Hidden Markov Model and do POS tagging for given statements

b Describe in detail Centering Algorithm for reference resolution.

Q.5 **10 marks each**

a For a given grammar using CYK or CKY algorithm parse the statement

"The man read this book"

Rules:

S → NP VP	Det → *that* | *this* | *a* | *the*
S → Aux NP VP	Noun → *book* | *flight* | *meal* | *man*
S → VP	Verb → *book* | *include* | *read*
NP → Det NOM	Aux → *does*
NOM → Noun	
NOM → Noun NOM	
VP → Verb	
VP → Verb NP	

b Explain Porter Stemmer algorithm with rules

Q.6 **10 marks each**

a Explain information retrieval versus Information extraction systems

b Explain Maximum Entropy Model for POS Tagging

Duration: 3hrs **[Max Marks: 80]**

N.B.: (1) Question No 1 is Compulsory.
(2) Attempt any three questions out of the remaining five.
(3) All questions carry equal marks.
(4) Assume suitable data, if required and state it clearly.

1		Attempt any FOUR	**[20]**
	a	What is the rule-based and stochastic part of speech taggers?	
	b	Explain Good Turing Discounting?	
	c	Explain statistical approach for machine translation.	
	d	Explain with suitable example the following relationships between word meanings: Hyponymy, Hypernymy, Meronymy, Holynymy	
	e	What is reference resolution?	
2	**a**	Explain FSA for nouns and verbs. Also Design a Finite State Automata (FSA) for the words of English numbers 1-99.	**[10]**
	b	Discuss the challenges in various stages of natural language processing.	**[10]**
3	**a**	Consider the following corpus <s> the/DT students/NN pass/V the/DT test/NN<\s> <s> the/DT students/NN wait/V for/P the/DT result/NN<\s> <s> teachers/NN test/V students/NN<\s> Compute the emission and transition probabilities for a bigram HMM. Also decode the following sentence using Viterbi algorithm. **"The students wait for the test"**	**[10]**
	b	What are five types of referring expressions? Explain with the help of example.	**[10]**
4	**a**	Explain dictionary-based approach (Lesk algorithm) for word sense disambiguation (WSD) with suitable example.	**[10]**
	b	Explain the various challenges in POS tagging.	**[10]**
5	**a**	Explain Porter Stemming algorithm in detail.	**[10]**
	b	Explain the use of Probabilistic Context Free Grammar (PCFG) in natural language processing with example.	**[10]**
6	**a**	Explain Question Answering system (QAS) in detail.	**[10]**
	b	Explain how Conditional Random Field (CRF) is used for sequence labeling.	**[10]**

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Time: 3 Hours **Max. Marks: 80**

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N.B. (1) Question No. 1 is compulsory

(2) Assume suitable data if necessary

(3)Attempt any three questions from remaining questions

Q.1	Any Four	**20[M]**
a	Differentiate between Syntactic ambiguity and Lexical Ambiguity.	**[5M]**
b	Define affixes. Explain the types of affixes.	**[5M]**
c	Describe open class words and closed class words in English with examples.	**[5M]**
d	What is rule base machine translation?	**[5M]**
e	Explain with suitable example following relationships between word meanings. Homonymy, Polysemy, Synonymy, Antonymy	**[5M]**
f	Explain perplexity of any language model.	**[5M]**
Q.2 a)	Explain the role of FSA in morphological analysis?	
Q.2 b)	Explain Different stage involved in NLP process with suitable example.	**[10M]**
Q.3 a)	Consider the following corpus <s> I tell you to sleep and rest </s> <s> I would like to sleep for an hour </s> <s> Sleep helps one to relax </s> List all possible bigrams. Compute conditional probabilities and predict the next ord for the word “to”.	**[5M]**
Q.3 b)	Explain Yarowsky bootstrapping approach of semi supervised learning	**[5M]**
Q.3 c)	What is POS tagging? Discuss various challenges faced by POS tagging.	**[10M]**
Q.4 a)	What are the limitations of Hidden Markov Model?	**[5M]**
Q.4 b)	Explain the different steps in text processing for Information Retrieval	**[5M]**
Q.4 c)	Compare top-down and bottom-up approach of parsing with example.	**[10M]**
Q.5 a)	What do you mean by word sense disambiguation (WSD)? Discuss dictionary based approach for WSD.	**[10M]**
Q.5 b)	Explain Hobbs algorithm for pronoun resolution.	**[10M]**
Q.6 a)	Explain Text summarization in detail.	**[10M]**
Q.6 b)	Explain Porter Stemming algorithm in detail	**[10M]**

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